The Dimension of Nonterminating Resampling Computations

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Abstract

A randomized algorithm may terminate almost surely although exceptional random tapes make it run forever. Motivated by resampling algorithms for the Lovász local lemma, this paper studies three quantities: the probability of surviving T repairs, the algorithmic randomness of one nonterminating tape, and the Hausdorff dimension of all such tapes.

Two complementary reductions are developed. Reading one finite resampling run backward gives an exact likelihood ratio whose tail and coding forms give running-time bounds when the information surplus grows and description-length bounds for every surviving prefix. Summing source probabilities to a variable power gives a set-level pressure; a state statistic that preserves these transition weights at every power determines survival and dimension exactly.

For finite extremal partial rejection, this pressure places the nontermination dimension at a powered Shearer boundary. For nonextremal disagreement repair on any connected graph, Hamming weight determines the complete survival law and dimension independently of graph topology and every deterministic nonanticipating repair rule. The calculation uses $N-1$ pressure states, although the minimal exact-continuation machine may require 2^{N-1} states. Whole-block examples further show that dependency graphs, event marginals, and even all pairwise intersection probabilities need not determine exact survival or dimension. A comparable small exact state for arbitrary overlapping repairs remains open.

Contents

1 Introduction

A randomized algorithm is deterministic once its random tape is fixed. Although the algorithm may terminate with probability one, some tapes may make it run forever. Let $\mathbb{P}$ denote the source law of the tape and put

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where τ is the number of rejection or repair steps before stopping. The tail $\mathbb{P}\{\tau \geq T\}$ measures how often the computation survives T steps. When the following extended limit exists, the *survival exponent* (or base-two escape rate) is

$$\alpha = -\lim_{T\to\infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

This terminology is standard for exponential escape from a surviving region (Keller and Liverani, 2009). This distributional quantity does not describe the exceptional tapes themselves. For a computable process, $\mathcal{N}$ is effectively closed because termination is witnessed by a finite tape prefix. Hence a P-null $\mathcal{N}$ contains no P–Martin–Löf-random tape. It may nevertheless have positive or even full Hausdorff dimension and contain tapes of maximal effective strong dimension. Effective dimension measures the algorithmic randomness of one fixed tape; source-relative Hausdorff dimension measures the size of the whole exceptional set.

The Lovász local lemma has a different scope. It gives conditions under which all bad events can be avoided, and its algorithmic forms give termination or tail guarantees from event probabilities and dependency structure (Erdős and Lovász, 1975; Moser and Tardos, 2010; Haeupler and Harris, 2017). The present paper fixes the resampling computation and asks for its exact survival decay and for the complexity and geometry of its exceptional tapes. These are questions about a specified process, not a stronger universal local-lemma criterion.

1.1 Pointwise complexity and localization

In statistical learning, $\widehat{h}(S)$ depends on the training sample S, so a bound for one fixed hypothesis cannot simply be evaluated at the learned one. Uniform and localized empirical-process methods instead control the data-dependent region in which the estimator lies (Bartlett et al., 2005; Koltchinskii, 2006; Xu and Zeevi, 2025). A resampling computation has the analogous dependence between a random tape and the repair history it generates: $\{\tau \geq T\}$ restricts tape space to the nested family of live prefixes selected by the computation. Recent work localizes complexity at one trained neural network or one index of a random field (Li and Xu, 2026; Xu, 2026b); here the pointwise object is one random tape. The later search for a smaller state preserving specified quantities also parallels decision-sufficient state compression (Xu, 2026a). These parallels motivate the questions but are not reductions: local Rademacher complexity, Kolmogorov complexity, and weighted cylinder sums are distinct objects, and no learning-theoretic generalization bound is claimed.

For a computable full-support source $P = \pi^{\mathbb{N}}$, let $\omega_{1:n}$ denote the first n source symbols (or blocks). For a finite word w, let $[w]$ be the cylinder of tapes beginning with w, and write $P[w] = P([w])$. Its self-information is

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}].$$

Let $K(w)$ be the length of the shortest program that prints w on a fixed universal prefix-free, or self-delimiting, machine (Li and Vitányi, 2019). More generally, $K(w \mid z)$ is the corresponding program length when z is supplied as auxiliary input. The effective strong P-dimension of one tape is

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) = \limsup_{n\to\infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \tag{1}$$

Replacing the limsup by a liminf defines the effective P-dimension $\dim_{\mathrm{eff},P}(\omega)$. For a fair binary source, write

$$\mathrm{Dim}_{\mathrm{eff}}(\omega) = \limsup_{n\to\infty} \frac{K(\omega_{1:n})}{n}.$$

These are the standard constructive dimensions relative to a source measure (Athreya et al., 2007; Lutz, 2011); the definitions themselves are not new. A value near one means that the tape retains nearly the full algorithmic information of the source, while a smaller value means that continued execution has imposed additional structure.

For a set E of tapes, $\dim_H^P E$ denotes its Billingsley, or source-relative Hausdorff, dimension: in the Hausdorff covering construction, the cylinder $[w]$ has diameter $P[w]$ (Billingsley, 1965; Lutz, 2011). Under a uniform q-ary source, this is ordinary Hausdorff dimension in the symbolic metric $d(\omega, \eta) = q^{-k}$, where k is the length of the longest common prefix. Section 2.1 gives the corresponding source tree and metric when the amount of tape consumed is adaptive.

1.2 Two complementary principles

The paper uses two principles with different quantifiers. The first is pathwise. Comparing one finite run with any subprobability law Q_T on feasible repair histories gives the information surplus

$$\Delta_T = \sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(A_1, \ldots, A_T)}.$$

Reading the run backward shows that this is an exact log-likelihood ratio. Consequently $\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}$; under computability, the same backward law describes each surviving prefix using Δ_T fewer bits than its source self-information, up to an additive constant. This identity requires no local-lemma hypothesis.

The second principle is setwise. For the prefix-free family $\mathcal{L}_T$ of consumed tape prefixes that remain live for T repairs, define

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s.$$

This is the Rényi partition sum of the live cylinders, whose exponential growth is pressure (Rényi, 1961; Bowen, 1975). At $s = 1$ it is the survival probability; its critical power gives Hausdorff dimension. For a finite-state computation it is governed by a nonnegative operator B_s. If a statistic preserves the total $p(e)^s$-weight of transitions between its classes for every s, a reduced matrix R_s preserves $Z_T(s)$ exactly. Thus $\rho(R_1)$ gives the survival rate and the solution of $\rho(R_\delta) = 1$ gives the dimension.

These principles are complementary. The likelihood identity concerns one realized run and needs no finite-state description. Pressure sums over all surviving runs and gives exact asymptotic answers when their weighted growth has a tractable representation. Neither is a consequence of the other.

1.3 Main results

The principal results are as follows.

1. Theorem 2.1 is a finite-run identity for every variable-model resampling computation. It yields a universal tail bound, an individual-tape coding bound, and an exact relative-entropy expression for the slack.

2. Source-power analysis is exact by two different mechanisms. Extremal repair admits Cartier–Foata trace enumeration, whereas Proposition 4.2 reduces every powered transition operator for nonextremal disagreement repair.

3. Theorem 5.1 computes the survival exponent and both tape dimensions for every finite extremal partial-rejection instance. Powering each consumed source value lifts the classical Cartier–Foata trace series to the complete source-power family; the nontermination dimension is the value at which the source-power event masses reach the Shearer boundary.

4. Theorem 5.4 solves a genuinely nonextremal family. For disagreement repair on any connected graph, Hamming weight preserves every source-power sum, independently of the deterministic nonanticipating repair rule. On a path under the leftmost rule, the resulting pressure calculation uses $N - 1$ states, whereas the minimal exact-continuation machine has at least 2^{N-1} states. This is an exponential separation in state cardinality, not an exponential-space lower bound.

Whole-block systems provide a limitation rather than a stronger LLL theorem: the same event–variable graph, dependency graph, event marginals, and all pairwise intersection probabilities can give different survival exponents and dimensions because these summaries omit higher-order intersections.

1.4 Relation to earlier work and scope

Reverse reconstruction is an established entropy-compression device (Moser, 2009; Moser and Tardos, 2010). In particular, Messner and Thierauf (2012) reconstruct the uniform random bits used by

Moser's satisfiability search from the final assignment and a compressed repair log, and then apply Kolmogorov complexity. In the broader framework of focused stochastic search, Achlioptas et al. (2019, Theorem 5) give sharp product-form probabilities for feasible flaw sequences under atomicity and regeneration. The backward theorem here does not claim reconstruction as new. Its contribution is an exact likelihood formulation for arbitrary finite product sources and non-atomic variable-model events, the freedom to choose Q_T, and the resulting KL slack and pointwise coding bound.

For extremal partial rejection, confluence, Cartier–Foata traces, and the exact probability-weighted transcript series are known (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024). Abbes and Mairesse (2015) characterize Bernoulli measures on irreducible infinite trace boundaries through Möbius vanishing and positivity and realize them as Markov chains on Cartier–Foata cliques. Thus neither trace enumeration nor the existence of a critical trace-boundary law is claimed as new. The additional step is to power each consumed source value before aggregation, replacing p_i^s by its within-event Rényi mass, and to identify the resulting critical power with the source-relative Hausdorff and maximal effective strong dimensions of the consumed tapes. For reducible dependency graphs, the analysis is componentwise.

The finite-state connection among weighted growth, Perron eigenvectors, and Hausdorff dimension is classical (Parry, 1964; Bowen, 1975; Mauldin and Williams, 1988; Simpson, 2015). At $s = 1$ the reduction below is ordinary Markov lumpability, closely related to probabilistic bisimulation (Kemeny and Snell, 1960; Larsen and Skou, 1991); its all-power form is naturally compared with weighted stochastic automata (Carlyle and Paz, 1971). The continuation lower bounds are Myhill–Nerode-style distinctions. Probabilistic-program verification usually studies reachability, almost-sure termination, or expected time (Etessami and Yannakakis, 2009; Fioriti and Hermanns, 2015); the present question concerns the individual complexity and Hausdorff geometry of the exceptional tapes of a fixed computation.

The paper does not give a new universal LLL threshold, a general polynomial-time algorithm for exact dimension, or a compact exact state for arbitrary overlapping repairs. When the selector is determined by the current assignment, that assignment gives an exact finite-state operator; finite controller memory can be adjoined, but the resulting state space is usually exponential. A broader compilation or hardness theorem for succinctly described overlapping repairs remains the main open direction. The finite-run identity is presented first; rejection sampling appears only afterward as a one-state calibration. All proofs are collected in the appendices.

2 A backward likelihood identity

Let $X_1, \dots, X_n$ be independent variables, with X_i taking values in a finite set $\mathcal{X}_i$, and with product law $\pi = \bigotimes_i \pi_i$. For $J \subseteq \{1, \dots, n\}$, write $\pi_J = \bigotimes_{i \in J} \pi_i$ for the marginal product law on the coordinates in J. Each nonempty bad event A is determined by the coordinates $v(A)$ and has probability $p_A > 0$. Its *marginal surprisal* is $\log_2(1/p_A)$. Fix a deterministic rule for selecting a currently true bad event.

List the random choices in the order in which the algorithm uses them. First draw $S_0 \sim \pi$. Whenever the current assignment contains a bad event, select A_t, draw $Z_t \sim \pi_{v(A_t)}$ independently of the past, and write Z_t on $v(A_t)$. Let S_t denote the resulting assignment after the tth repair. Let E_T be the event that at least T resamplings occur, and put

$$\Xi_T = (S_0, Z_1, \dots, Z_T).$$

2.1 The consumed-tape space

The tape is the rooted tree of source choices actually queried by the computation. A finite node is a feasible transcript

$$w = (s_0, z_1, \dots, z_t).$$

The root branches to s_0 with probability $\pi(s_0)$. If w is live and the rule next selects A_{t+1}, its children are indexed by $z \in \prod_{i \in v(A_{t+1})} \mathcal{X}_i$, with edge probability $\pi_{v(A_{t+1})}(z)$. Thus the cylinder rooted at w has mass

$$P_{\mathrm{tape}}[w] = \pi(s_0) \prod_{j=1}^{t} \pi_{v(A_j)}(z_j). \tag{2}$$

Terminal transcripts are leaves, and infinite branches are precisely the nonterminating consumed tapes. For distinct infinite branches ω and η, let $\omega \wedge \eta$ be their longest common consumed prefix and set

$$d_P(\omega, \eta) = P_{\mathrm{tape}}[\omega \wedge \eta]. \tag{3}$$

Equivalently, source-relative Hausdorff dimension uses the cylinder gauge $P_{\mathrm{tape}}[w]$. All dimension results below assume that these masses tend to zero along every infinite branch.

Fix computable enumerations of the finite outcome alphabets. They give a canonical self-delimiting encoding of every transcript; the selected scope is recovered from the preceding transcript and the deterministic rule. The notation $K(w)$ refers to this encoding. The dimensions are unchanged under a computable prefix-tree isomorphism Ψ with computable inverse and bounded distortion

$$C^{-1} P_{\mathrm{tape}}[w] \le P'_{\mathrm{tape}}[\Psi(w)] \le C P_{\mathrm{tape}}[w]$$

for some fixed C: the two self-informations and the two prefix complexities then differ by $O(1)$. No invariance under arbitrary recodings is claimed.

A pre-sampled resampling table is only a coupling device. Entries not read by the realized computation are excluded: adjoining independent unused coordinates can change a tape dimension without changing either the computation or its stopping time.

Let P_{tape} denote this sequential source law, and abbreviate $P_{\mathrm{tape}}(w) = P_{\mathrm{tape}}[w]$. Thus $E_T = \{\tau \ge T\}$. The minimal consumed prefixes that realize T repairs form a prefix-free family, so their cylinders are disjoint and their probabilities sum to $\mathbb{P}(E_T)$.

Let $Y_t = S_{t-1}|_{v(A_t)}$ be the configuration overwritten at time t, and let $L_T = (A_1, \dots, A_T)$. Write $\pi_{v(A)}(\cdot \mid A)$ for the marginal law $\pi_{v(A)}$ conditioned on the local configurations satisfying A. Coordinatewise, the entries of the used tape are partitioned into the final assignment and the overwritten configurations. Therefore

$$P_{\mathrm{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^{T} p_{A_t}\, \pi_{v(A_t)}(Y_t \mid A_t). \tag{4}$$

Let Q_T be a subprobability law on repair histories $\ell = (A_1, \dots, A_T)$, positive on every feasible history of length T. Thus $-\log_2 Q_T(\ell)$ is its ideal code length. Define the backward subprobability law on complete records by

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^{T} \pi_{v(A_t)}(y_t \mid A_t). \tag{5}$$

The map

$$\Xi_T \longmapsto (S_T, L_T, Y_{1:T})$$

is injective. Indeed, reverse the repairs from T to 1: the current values on $v(A_t)$ recover Z_t, and restoring Y_t recovers the preceding assignment.

Reverse reconstruction itself is an established entropy-compression device. In particular, Messner and Thierauf (2012) reconstruct the uniform random bits used by Moser's satisfiability search from its repair log and final assignment, and then apply Kolmogorov complexity. The result below adds an exact likelihood formulation for nonuniform sources and non-atomic events, an arbitrary comparison law Q_T, and a common tail, KL, and coding interpretation.

Theorem 2.1 (Backward likelihood identity)**.** *Every execution in E_T satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\mathrm{tape}}(\Xi_T)} = \sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \tag{6}$$

Call the right side Δ_T on E_T and set $\Delta_T = -\infty$ otherwise. Then, for every $u \geq 0$,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \tag{7}$$

If the instance and the family $(Q_T)_{T\geq 1}$ are uniformly computable, then

$$-\log_2 P_{\mathrm{tape}}(\Xi_T) - K(\Xi_T \mid T, \mathrm{instance}) \geq \Delta_T - O(1). \tag{8}$$

Here instance *denotes a fixed effective description of the source laws, bad events, and deterministic selection rule. Uniform computability means that one algorithm can approximate $Q_T(\ell)$ to any requested precision from T and ℓ, with the instance data encoded effectively in the same sense.*

For atomic regenerating focused stochastic search, Achlioptas et al. (2019, Theorem 5) obtain sharp product-form probabilities for feasible flaw sequences. The present record retains the overwritten local configurations, so the identity also applies to non-atomic variable-model events; the arbitrary law Q_T then exposes both the exact mismatch in (9) below and the coding consequence (8).

For use in the exact slack formula below, let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes producing at least T repairs and write

$$R_T(w) = \big(S_T(w), L_T(w), Y_{1:T}(w)\big), \qquad \mu_T(w) = G_T(R_T(w)).$$

Backward reconstruction makes R_T injective. Since G_T has total mass at most one, μ_T is a subprobability mass on $\mathcal{W}_T$.

The proof is given in Appendix C.

The likelihood identity also identifies the exact slack after conditioning on a long run. Let $p_T = P_{\mathrm{tape}}(E_T) > 0$, let $m_T = \sum_{w\in\mathcal{W}_T} \mu_T(w) > 0$, and define probability laws on $\mathcal{W}_T$ by

$$P_T(w) = \frac{P_{\mathrm{tape}}(w)}{p_T}, \qquad \nu_T(w) = \frac{\mu_T(w)}{m_T}.$$

Write D_2 for relative entropy in bits and H_2 for Shannon entropy in bits.

Corollary 2.2 (Exact slack and the optimal history law)**.** *For every fixed comparison law Q_T,*

$$-\log_2 p_T = \mathbb{E}_{P_T}\Delta_T + D_2(P_T \| \nu_T) - \log_2 m_T. \tag{9}$$

Moreover, if $J_T = \sum_{t=1}^{T} \log_2(1/p_{A_t})$, then

$$\sup_{Q_T} \mathbb{E}[\Delta_T \mid E_T] = \mathbb{E}[J_T \mid E_T] - H_2(L_T \mid E_T), \tag{10}$$

where the supremum is over subprobability laws positive on feasible histories. It is attained by $Q_T(\ell) = \mathbb{P}\{L_T = \ell \mid E_T\}$ on the conditional support.

The proof is given in Appendix C.

Thus the difference between the escape information $-\log_2 p_T$ and the mean surplus consists exactly of a mismatch term and the unused backward mass. The theorem also gives the normalized bound

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \geq a\right\} \leq 2^{-aT} \qquad (T \geq 1,\ a \geq 0). \tag{11}$$

Corollary 2.3 (Sharpness of the surplus tail)**.** *Neither the constant nor the exponent in (11) can be improved universally. For any $a > 0$, a one-event rejection sampler with event probability $p = 2^{-a}$ and its unique repair history assigned probability one satisfies, for every T,*

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \geq a\right\} = \mathbb{P}(E_T) = p^T = 2^{-aT}. \tag{12}$$

Large-deviation interpretation. No large-deviation principle is assumed in the paper. If Δ_T/T is known separately to satisfy one with a base-two rate function J, and the principle identifies the exact upper-tail rate at a, then

$$\inf_{x \geq a} J(x) \geq a. \tag{13}$$

The one-event example shows that equality is possible. For ordinary rejection sampling with rejection probability p, survival means that T independent Bernoulli variables all equal one, so its rate is the endpoint relative entropy

$$D_2(1\|p) = \log_2 \frac{1}{p}.$$

For the finite-state model in Appendix A, the corresponding escape rate is $-\log_2 \rho(B_1)$ (Dembo and Zeitouni, 2010; Keller and Liverani, 2009).

The two terms in (6) have different origins. The first sums the marginal surprisals of the events that were repaired; the second is the ideal code length of their ordered history. The first sum may count statistically dependent information more than once. Neither term alone gives a bound. Their difference does, because it is a likelihood ratio.

For example, color the vertices of a graph independently with q colors and repeatedly recolor the endpoints of the first monochromatic edge. A fixed edge is monochromatic with probability $1/q$, so each repair contributes $\log_2 q$ bits to the first term. The second term pays for the repaired edges and their order. The comparison is therefore between information ruled out by the violations and information used by the algorithm's history.

Corollary 2.4 (A tail bound and a pointwise dimension bound)**.** *Suppose every T-repair execution satisfies*

$$\sum_{t=1}^{T} \log_2 \frac{1}{p_{A_t}} \geq \beta T - C_0, \qquad -\log_2 Q_T(L_T) \leq \gamma T + C_1,$$

where $\beta > \gamma$. Then

$$\mathbb{P}(E_T) \leq 2^{C_0 + C_1 - (\beta - \gamma)T}. \tag{14}$$

For the pointwise conclusion, assume in addition that

(i) the instance and the family $(Q_T)_{T \geq 1}$ are uniformly computable;

(ii) the computation reads a fixed fair binary tape sequentially, the initial draw consumes $c_0 = O(1)$ bits, and every repair consumes exactly c bits, so that Ξ_T is uniformly computably interconvertible with $\omega_{1:c_0+cT}$.

Then every nonterminating tape satisfies

$$\mathrm{Dim}_{\mathrm{eff}}(\omega) \le \max\left\{0, 1 - \frac{\beta - \gamma}{c}\right\}. \tag{15}$$

The proof is given in Appendix C.

Remark 2.5 (What the record must contain). For a recursive Moser-style implementation, L_T must be enlarged to include the recursion tree or return structure unless those data are recoverable by a separate prefix-free decoding rule. In Theorem 2.1, everything needed for reversal contributes to the probability $Q_T(L_T)$.

3 A one-state calibration: rejection sampling

Consider the simplest deterministic form of rejection sampling. Draw $X_1, X_2, \dots$ independently from a full-support law π on a finite set Ω, reject a draw in $U \subseteq \Omega$, and return the first draw in $G = \Omega \setminus U$. This is a special case of the classical accept–reject method (von Neumann, 1951; Devroye, 1986). Write $p = \pi(U)$. If $0 < p < 1$ and τ is the number of rejected draws, then

$$\mathbb{P}\{\tau \ge n\} = p^n, \qquad \mathbb{P}\{\tau = t\} = p^t(1-p), \qquad \mathbb{E}\tau = \frac{p}{1-p}. \tag{16}$$

The returned value has law $\pi(\,\cdot \mid G)$. The algorithm terminates almost surely, but its nonterminating set is the nonempty null set $\mathcal{N} = U^{\mathbb{N}}$.

This one-state process calibrates the two notions of tape dimension. In general, effective strong dimension is associated with packing rather than Hausdorff dimension (Athreya et al., 2007); their equality here is a property of a regular product set, not a general identity. If U is nonempty, let $\delta \in [0,1]$ be the unique solution of

$$\sum_{x \in U} \pi(x)^{\delta} = 1. \tag{17}$$

Then

$$\sup_{\omega \in \mathcal{N}} \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \dim_H^P \mathcal{N} = \delta, \qquad \mathrm{Dim}_{\mathrm{eff},P}(\omega) \le \delta \quad (\omega \in \mathcal{N}). \tag{18}$$

Under computability the supremum is attained. Indeed, writing $\mathbf{1}\{\cdot\}$ for an indicator, the tilted law

$$Q_\delta(x) = \pi(x)^{\delta}\mathbf{1}\{x \in U\}$$

is a computable probability law, and every tape that is Martin–Löf random under $Q_\delta^{\mathbb{N}}$ attains δ by the Levin–Schnorr theorem (Li and Vitányi, 2019). Under a uniform q-ary source, $\pi(U) = q^{\delta-1}$, so the tail exponent and dimension determine one another; for a nonuniform source they need not.

For example, take $\Omega = \{00, 01, 10, 11\}$ uniformly and $U = \{00, 01\}$. Every surviving block has its first bit fixed and its second bit free, so both the maximal effective strong dimension and the Hausdorff dimension are $1/2$. Proposition B.1 in Appendix B gives the corresponding Rényi-entropy identity for a nonuniform source. These facts use standard product geometry and are not claimed as a new rejection-sampling theorem; their role is to fix, in one state, the survival and dimension normalizations used below.

4 Exact source-power state reduction

Let $\mathcal{X}$ be a finite set of live states. From $x \in \mathcal{X}$, a source outcome e of probability $p(e) > 0$ either moves the computation to a live state $F(x, e)$ or terminates it. For $s \geq 0$, let

$$(B_s)_{xy} = \sum_{e:F(x,e)=y} p(e)^s, \qquad x, y \in \mathcal{X}. \tag{19}$$

Thus B_1 is the usual killed transition matrix, whereas the family $(B_s)_{s\geq 0}$ counts source-weighted tape prefixes.

The following standard finite-state pressure formula is stated here because it is the bridge from the operators to the three quantities studied in the paper. Its detailed form and proof appear in Appendix A.

Proposition 4.1 (Finite-state pressure)**.** *Fix an initial live state. Suppose its reachable live-state graph is finite, strongly connected, contains a directed cycle, and satisfies* $\max_e p(e) < 1$. *There is a unique* $\delta \in [0, 1]$ *such that* $\rho(B_\delta) = 1$. *The survival exponent is* $-\log_2 \rho(B_1)$, *and the source-relative Hausdorff dimension of the nonterminating tapes is* δ. *If the system and its critical Perron law are computable, every nonterminating tape has effective strong dimension at most* δ, *and some tape attains* δ. *For a reducible graph, these statements apply componentwise to reachable cyclic strongly connected components.*

Let $\phi : \mathcal{X} \to \mathcal{C}$ be a surjection. Say that ϕ *lumps the source powers* if, for every $s \geq 0$ and $c, d \in \mathcal{C}$, the quantity

$$R_s(c, d) := \sum_{e:F(x,e)\in\phi^{-1}(d)} p(e)^s \tag{20}$$

does not depend on the choice of $x \in \phi^{-1}(c)$. This phrase is descriptive rather than standard. At $s = 1$, condition (20) is the usual strong lumpability condition for a Markov chain (Kemeny and Snell, 1960); requiring it for all s is what also preserves the geometry of the source tape.

Proposition 4.2 (Exact source-power reduction)**.** *If* ϕ *satisfies* (20), *then, for every* $x \in \mathcal{X}$ *and integer* $T \geq 0$,

$$(B_s^T \mathbf{1})(x) = (R_s^T \mathbf{1})(\phi(x)). \tag{21}$$

In particular,

$$\rho(B_s) = \rho(R_s). \tag{22}$$

If $R_s r_s = \varrho_s r_s$ *for a strictly positive* r_s *and* $\varrho_s > 0$, *then* $h_s = r_s \circ \phi$ *satisfies* $B_s h_s = \varrho_s h_s$, *and*

$$Q_s(e \mid x) = \frac{p(e)^s h_s(F(x, e))}{\varrho_s h_s(x)} \tag{23}$$

for outcomes with $F(x, e) \in \mathcal{X}$ *(and zero otherwise) is a probability law on live transitions. Every path* $w = e_1 \cdots e_T$ *from* x_0 *to* x_T *then obeys*

$$\log_2 \frac{Q_s[w]}{P[w]} = (1 - s) \log_2 \frac{1}{P[w]} - T \log_2 \varrho_s + \log_2 \frac{h_s(x_T)}{h_s(x_0)}. \tag{24}$$

Consequently, under the hypotheses of Proposition 4.1, the survival exponent is $-\log_2 \rho(R_1)$ *and the nontermination dimension is the solution of* $\rho(R_s) = 1$. *The same reduced eigenfunction gives the dimension-attaining Doob transform and the effective-dimension bound.*

The proof is given in Appendix E.

The continuum of powers in (20) is not an algorithmic obstacle when the transition system is explicit. Let $\mathcal{R}$ be the finite set of values taken by $p(e)$. Distinct functions r^s, $r \in \mathcal{R}$, are linearly independent as functions of s. Hence (20) is equivalent to requiring, for each $r \in \mathcal{R}$, that the number of probability-r outcomes from x into each class d depend only on $\phi(x)$. When the probabilities are given by exact labels with decidable equality—for example, as rational or algebraic numbers—ordinary labelled partition refinement therefore finds the coarsest source-power lumping in time polynomial in the *explicit* transition graph. This does not by itself avoid an exponentially large assignment graph; it identifies exactly what a smaller representation must preserve.

There is a simple local criterion for Proposition 4.2. Suppose an assignment has an additive statistic $g(x) = \sum_v g_v(x_v)$ and that terminal status depends only on $g(x)$. If, whenever $g(x) = k$, the selected repair block has a fixed old score $c(k)$ and the total s-weight of its redraws with new score j is a number $\nu_{s,k}(j)$ independent of x, then g lumps the source powers, with

$$R_s\big(k, k - c(k) + j\big) = \nu_{s,k}(j). \tag{25}$$

This criterion allows the selected block itself to depend on the full assignment. Only its contribution to the statistic, and the source-weighted law of its replacement, must be determined by k.

5 Exact local repair

Two mechanisms give exact source-power sums below. Extremal repair is enumerated by commuting-event traces; disagreement repair admits the all-power state reduction of Proposition 4.2. The second case also permits a direct comparison between the state needed for survival and dimension and the state needed to continue the computation exactly.

5.1 Extremal local repair and the Shearer boundary

The next result computes survival and both tape dimensions for every finite extremal instance of partial rejection sampling. Let $(X_v)_{v\in V}$ be independent finite-valued variables with full-support laws π_v. There is at least one bad event, and A_i depends on the scope $V_i \subseteq V$ and is represented by a set $\mathcal{B}_i \subseteq \prod_{v\in V_i} \mathcal{X}_v$ of bad local configurations. The dependency graph D has vertex set $[m]$ and joins i and j when $V_i \cap V_j \neq \varnothing$. The instance is *extremal* if

$$A_i \cap A_j = \varnothing \qquad (ij \in E(D)).$$

Thus two events are either supported on disjoint variables or cannot occur simultaneously. Assume that the satisfying set $\Sigma = \bigcap_i A_i^c$ is nonempty and that every A_i is nonempty. At each step, a fixed deterministic current-assignment rule selects a currently true event and redraws the variables in its scope. Let P be the resulting sequential source law. Assume also that every bad local configuration has source probability strictly between zero and one. Only the values actually drawn, in their order of revelation, form the tape. This is the consumed-choice tree of Section 2.1; unused entries of a resampling table are not additional tape coordinates.

For $s \geq 0$, define the source-power mass of event i and of the satisfying frontier by

$$r_i(s) = \sum_{b\in\mathcal{B}_i} \pi_{V_i}(b)^s, \qquad Z_\Sigma(s) = \sum_{x\in\Sigma} \pi(x)^s. \tag{26}$$

In particular, $r_i(1) = p_i := \mathbb{P}(A_i)$. If A_i is atomic, meaning that $\mathcal{B}_i$ is a singleton, then $r_i(s) = p_i^s$.

Let

$$P_D(\mathbf{z}) = \sum_{I\subseteq[m]\,:\,I \text{ independent in } D} (-1)^{|I|} \prod_{i\in I} z_i \tag{27}$$

be the signed independence polynomial. For this graph, Shearer's region is the downward-closed set of vectors $\mathbf{z} \geq 0$ such that $P_D(\mathbf{y}) > 0$ for every $0 \leq \mathbf{y} \leq \mathbf{z}$; its boundary is where this positivity first fails along a positive ray (Shearer, 1985; Scott and Sokal, 2005). In the trace monoid $\mathcal{M}(D)$, letters i and j commute exactly when $ij \notin E(D)$. Write $N_i(h)$ for the number of occurrences of i in a trace h, and $|h| = \sum_i N_i(h)$. Set

$$a_T(s) = \sum_{\substack{h\in\mathcal{M}(D)\\ |h|=T}} \prod_{i=1}^{m} r_i(s)^{N_i(h)}. \tag{28}$$

Theorem 5.1 (Extremal source-power formula)**.** *Let $\mathcal{H}_T$ be the set of minimal consumed tape prefixes ending when the algorithm halts after exactly T repairs. Then, as an identity of formal power series,*

$$\sum_{w\in\mathcal{H}_T} P[w]^s = Z_\Sigma(s) a_T(s), \tag{29}$$

$$\sum_{T\geq 0} u^T \sum_{w\in\mathcal{H}_T} P[w]^s = \frac{Z_\Sigma(s)}{P_D\big(ur_1(s), \ldots, ur_m(s)\big)}. \tag{30}$$

Let $\mathcal{L}_T$ be the set of prefixes for which the computation is still live after T repairs and put $L_T(s) = \sum_{w\in\mathcal{L}_T} P[w]^s$. For every fixed $s \geq 0$ there are constants $0 < c_s \leq C_s < \infty$, independent of T, such that

$$c_s a_T(s) \leq L_T(s) \leq C_s a_T(s), \qquad T \geq 1. \tag{31}$$

Here $L_T(1) = \mathbb{P}\{\tau > T\}$. Replacing T by $T-1$ gives the convention $\mathbb{P}\{\tau \geq T\}$ used for the survival exponent; this shift changes neither an exponential rate nor a dimension.

Let $R(s)$ be the radius of convergence in u of the right-hand side of (30). Equivalently, $R(s)$ is the first positive zero of $u \mapsto P_D(u\mathbf{r}(s))$. There is a unique $\delta \in [0, 1]$ such that $R(\delta) = 1$, and

$$\lim_{T\to\infty} \mathbb{P}\{\tau \geq T\}^{1/T} = R(1)^{-1}, \qquad \dim_H^P \mathcal{N} = \delta. \tag{32}$$

Moreover, as a formal power-series identity, and analytically for $|u| < R(1)$,

$$\mathbb{E}u^\tau = \frac{P_D(\mathbf{p})}{P_D(u\mathbf{p})}. \tag{33}$$

The vector $\mathbf{r}(\delta)$ lies on the boundary of Shearer's region for D; more explicitly,

$$P_D(\mathbf{r}(\delta)) = 0, \qquad P_D(u\mathbf{r}(\delta)) > 0 \quad (0 \leq u < 1). \tag{34}$$

If the source, events, and selection rule are computable, the largest effective strong dimension of a nonterminating tape is also δ, and this value is attained.

The proof is given in Appendix F.

The exact source-power masses also give a simple comparison using coarser event data. For a positive vector $\mathbf{z}$, let $\mathcal{R}_D(\mathbf{z})$ denote the first positive zero of $u \mapsto P_D(u\mathbf{z})$.

Corollary 5.2 (Marginal–multiplicity bounds)**.** *Put $k_i = |\mathcal{B}_i|$. Let δ_- and δ_+ be the unique solutions in $[0,1]$ of*

$$\mathcal{R}_D\big((p_i^{\delta_-})_i\big) = 1, \qquad \mathcal{R}_D\big((k_i^{1-\delta_+} p_i^{\delta_+})_i\big) = 1. \tag{35}$$

Then

$$\delta_- \le \dim_H^P \mathcal{N} \le \delta_+. \tag{36}$$

Let

$$\kappa_D = \limsup_{T\to\infty} \big|\{h \in \mathcal{M}(D) : |h| = T\}\big|^{1/T}$$

be the unweighted trace-growth rate. If all $p_i = p$ and all $k_i = k$, this becomes

$$\frac{\log_2 \kappa_D}{\log_2(1/p)} \le \dim_H^P \mathcal{N} \le \frac{\log_2(k\kappa_D)}{\log_2(k/p)}. \tag{37}$$

Both bounds coincide when the events are atomic.

The proof is given in Appendix F.

For atomic events, Theorem 5.1 takes a particularly simple form.

Corollary 5.3 (Powered-marginal Shearer boundary)**.** *If every bad event is atomic, then*

$$\dim_H^P \mathcal{N} = \delta, \qquad \mathbf{p}^{\delta} = (p_1^{\delta}, \dots, p_m^{\delta}) \text{ lies on the Shearer boundary of } D. \tag{38}$$

The same δ is the largest effective strong dimension under computability. If, in addition, $p_i = p$ for every i, put

$$\alpha = -\lim_{T\to\infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \ge T\}.$$

Then

$$\mathbb{P}\{\tau \ge T\}^{1/T} \longrightarrow p\kappa_D, \qquad \delta = \frac{\log_2 \kappa_D}{\log_2(1/p)} = 1 - \frac{\alpha}{\log_2(1/p)}. \tag{39}$$

For unequal atomic probabilities, with $\ell_{\min} = \min_i \log_2(1/p_i)$ and $\ell_{\max} = \max_i \log_2(1/p_i)$,

$$1 - \frac{\alpha}{\ell_{\min}} \le \delta \le 1 - \frac{\alpha}{\ell_{\max}}. \tag{40}$$

The proof is given in Appendix F.

Relation to classical trace enumeration. The confluence of extremal partial rejection sampling, the correspondence between transcripts and Cartier–Foata traces, and the identity $1/P_D$ are classical (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024). Jerrum's exact probability-weighted transcript series gives Equation (33) directly (Jerrum, 2024). Moreover, Abbes and Mairesse (2015) construct Bernoulli measures on irreducible infinite trace boundaries from Möbius vanishing and positivity. The additional step here is to power every consumed source value before summing and to treat reducible graphs componentwise. This identifies the source-power pressure, and hence the Hausdorff and pointwise dimensions of nontermination, with the point at which $\mathbf{r}(s)$ reaches the same boundary mechanism. For non-atomic events it is important to use $r_i(s)$ from (26); replacing it by p_i^s is generally incorrect (indeed $r_i(s) \ge p_i^s$ for $0 \le s \le 1$).

5.2 Repairing disagreements on a graph

Let $G = (V, E)$ be any connected simple graph with $N = |V| \geq 2$. Draw independent $X_v \sim$ Bernoulli(p), where $0 < p < 1$, and put $q = 1 - p$. For each edge $e = \{u, v\}$, define

$$A_e = \{X_u \neq X_v\}.$$

Fix any deterministic nonanticipating rule that selects a true event; the rule may use the current assignment and the past repair history, but not future source outcomes. To repair A_e, redraw X_u and X_v independently from their source laws. The terminal assignments are 0^V and 1^V. When G has two adjacent edges, the repairs genuinely overlap. For two distinct edges sharing one endpoint,

$$\mathbb{P}(A_e) = 2pq, \qquad \mathbb{P}(A_e \cap A_f) = pq. \tag{41}$$

Let τ be the number of repairs before a constant assignment is reached, with $\tau = 0$ if the initial assignment is constant. Let P be the sequential source law of the initial assignment and all redraws, let $\mathcal{N}(G, p) = \{\tau = \infty\}$, and for a redraw word $w = z_1 \cdots z_T$ write $P_p[w] = \prod_{t=1}^{T} P_p(z_t)$, where $P_p(z) = p^{z_1+z_2} q^{2-z_1-z_2}$ for $z = (z_1, z_2)$. For $s \geq 0$, define the $(N-1) \times (N-1)$ tridiagonal matrix R_s on $\{1, \ldots, N-1\}$ by

$$R_s(k, k-1) = q^{2s}, \qquad R_s(k, k) = 2(pq)^s, \qquad R_s(k, k+1) = p^{2s}, \tag{42}$$

where entries outside $\{1, \ldots, N-1\}$ are omitted.

Theorem 5.4 (Exact nontermination for disagreement repair)**.** *For every connected G, every deterministic nonanticipating selection rule, and every integer $T \geq 0$, the entire law of τ is the same. More precisely,*

$$\mathbb{P}\{\tau > T\} = \sum_{k=1}^{N-1} \binom{N}{k} p^k q^{N-k} (R_1^T \mathbf{1})_k. \tag{43}$$

More generally, the source-power sum of all tapes whose initial assignment and next T redraws remain live is

$$Z_T(s) = \sum_{k=1}^{N-1} \binom{N}{k} p^{sk} q^{s(N-k)} (R_s^T \mathbf{1})_k. \tag{44}$$

Thus $Z_T(1) = \mathbb{P}\{\tau > T\}$. Replacing T by $T-1$ recovers the convention $\mathbb{P}\{\tau \geq T\}$; this shift does not affect any exponential rate or dimension.

Set

$$\vartheta = \frac{\pi}{N}, \qquad c_N = \cos\left(\frac{\pi}{2N}\right).$$

The exact source-power rate, survival rate, and source-relative dimension are

$$\varrho_s = \lim_{T\to\infty} Z_T(s)^{1/T} = \rho(R_s) = 4(pq)^s c_N^2, \tag{45}$$

$$\lambda_{N,p} = \lim_{T\to\infty} \mathbb{P}\{\tau > T\}^{1/T} = 4pq\, c_N^2, \tag{46}$$

$$\dim_H^P \mathcal{N}(G, p) = \delta_{N,p} = \frac{\log(4c_N^2)}{\log(1/(pq))}. \tag{47}$$

For computable data, including computable p and a computable selection rule, $\delta_{N,p}$ is also the largest effective strong dimension of a nonterminating tape, and it is attained.

The dimension-attaining Doob transform is explicit. Start from any fixed computable live assignment; this finite initial choice does not affect dimension. If the current assignment has k ones and a redraw $z \in \{00, 01, 10, 11\}$ has $j(z)$ ones, put $d(z) = j(z) - 1$ and

$$Q^*(z \mid k) = \frac{\sin((k + d(z))\vartheta)}{4c_N^2 \sin(k\vartheta)}, \tag{48}$$

with value zero when $k + d(z) \in \{0, N\}$. This conditional law for the four redraw outcomes is independent of p, the graph, and the selection rule. Its Hamming-weight marginal is reversible with stationary distribution

$$\nu^*(k) = \frac{2}{N} \sin^2(k\vartheta), \qquad 1 \le k < N. \tag{49}$$

For every $s \ge 0$ and every surviving path $w = z_1 \cdots z_T$ from k_0 to k_T,

$$\log_2 \frac{Q^*[w \mid k_0]}{P_p[w]} = (1 - s)I_p(w) - T \log_2 \varrho_s + \log_2 \frac{h_s(k_T)}{h_s(k_0)}, \tag{50}$$

where

$$h_s(k) = \left(\frac{q}{p}\right)^{sk} \sin(k\vartheta), \qquad I_p(w) = \log_2 \frac{1}{P_p[w]}. \tag{51}$$

Set $h_s(0) = h_s(N) = 0$ when a boundary value occurs in a recurrence. Thus the same change of measure gives the survival exponent at $s = 1$ and the dimension bound at $s = \delta_{N,p}$.

The proof is given in Appendix G.

There is a particularly simple information identity behind the theorem. For one redraw,

$$P_p(z) = pq \left(\frac{p}{q}\right)^{d(z)}.$$

Consequently every surviving path satisfies

$$I_p(w) = T \log_2 \frac{1}{pq} + (k_T - k_0) \log_2 \frac{q}{p}. \tag{52}$$

The second term is bounded independently of T. If $\alpha_{N,p} = -\log_2 \lambda_{N,p}$, then

$$\delta_{N,p} = 1 - \frac{\alpha_{N,p}}{\log_2(1/(pq))}. \tag{53}$$

Equivalently, if $c = \log_2(1/(pq))$ is the source information per redraw up to the bounded endpoint term and $h = \log_2(4c_N^2)$ is the surviving information rate, then $\alpha_{N,p} = c - h$ and $\delta_{N,p} = h/c$. Thus $\alpha_{N,p} = c(1 - \delta_{N,p})$.

For a fair source,

$$\lambda_{N,1/2} = c_N^2, \qquad \delta_{N,1/2} = 1 + \log_2 c_N, \qquad \delta_{N,1/2} = 1 - \frac{\alpha_{N,1/2}}{2}. \tag{54}$$

Moreover,

$$\alpha_{N,1/2} = \frac{\pi^2}{4N^2 \ln 2} + O(N^{-4}), \qquad 1 - \delta_{N,1/2} = \frac{\pi^2}{8N^2 \ln 2} + O(N^{-4}). \tag{55}$$

Still under the fair-source assumption $p = 1/2$, the mean absorption time is also exact:

$$\mathbb{E}[\tau \mid M_0 = k] = 2k(N - k), \qquad \mathbb{E}\tau = \frac{N(N-1)}{2}. \tag{56}$$

Thus the process terminates almost surely and has quadratic mean running time, while the dimension of its nonterminating tapes tends to one.

The same result has a standard constraint-satisfaction form. Let $G = (L \cup R, E)$ be connected and bipartite, draw $X_v \sim \mathrm{Bernoulli}(p)$ on L and $X_v \sim \mathrm{Bernoulli}(q)$ on R, and declare a monochromatic edge bad. Repairing a bad edge redraws its endpoints from their respective source laws. The change of variables

$$Y_v = X_v \quad (v \in L), \qquad Y_v = 1 - X_v \quad (v \in R)$$

turns this process into disagreement repair with i.i.d. Bernoulli(p) variables. Therefore Theorem 5.4 gives the exact time and tape geometry of local resampling for a proper two-colouring, for every connected bipartite graph and every deterministic repair rule.

5.3 A small counting state versus a large continuation state

The statistic M determines all the quantities in Theorem 5.4, but it does not determine the next edge selected by a specified rule. This distinction cannot be removed merely by assuming that the event–variable incidence graph has small pathwidth.

Two live assignments are *continuation-equivalent* if every common future redraw sequence produces the same repair labels and the same halting decision. Any self-contained deterministic state description must refine this equivalence. Conversely, its equivalence classes form the minimal deterministic continuation machine, by the Myhill–Nerode principle.

Proposition 5.5 (A pathwidth-one state-cardinality obstruction)**.** *Let G be the path on vertices $0, 1, \ldots, N-1$, and always repair the leftmost disagreeing edge. Suppose a self-contained deterministic state summary has two properties: its current value determines the next repair (or halt), and it updates exactly from that value and the realized redraw. Then it has at least 2^{N-1} states when the halting state is included, and at least $2^{N-1} - 1$ live states.*

The proof is given in Appendix G.

The incidence graph in Proposition 5.5 is itself a path and has pathwidth one. Hence bounded pathwidth alone does not imply a $2^{O(w)}\,\mathrm{poly}(N)$ bound on the number of exact continuation states. The proposition exhibits 2^{N-1} distinct continuation classes, so a binary encoding of the current class needs at least $N-1$ bits in the worst case. This is a state-cardinality lower bound, not an exponential-space lower bound, and it does not assert that the scalar survival exponent is computationally hard.

By contrast, R_s is an $(N-1)$-state weighted linear representation of the source-power survival series. Such a representation need not distinguish all deterministic continuation classes (Carlyle and Paz, 1971). A statistic sufficient for pressure can therefore be much smaller as a state set than the minimal machine needed to reproduce every repair decision.

Relation to earlier processes. The reduction of a two-colour disagreement process to the number of vertices of one colour is closely related to gambler's-ruin reductions for discordant voting (Cooper et al., 2018). In that model a selected discordant edge is resolved by copying one endpoint, whereas here both endpoints are independently redrawn. The novelty claimed here is not the birth–death calculation by itself, but its simultaneous extension to every source-power operator, the all-power change-of-measure identity (50), the tape-dimension conclusions, and the separation between an exact pressure state and an exact continuation state. The instance is also nonextremal: adjacent bad events can occur together. Thus the extremal-instance analysis of partial rejection sampling does not yield this reduction (Guo et al., 2019; Jerrum, 2024).

A second extremal benchmark. Appendix I gives an adjacent-rise extremal family in which an $(n+1)$-vertex tridiagonal pressure matrix replaces 2^n exact continuation states. It is included as an application and benchmark for the source-power reduction, rather than as a separate principal theorem.

6 What graph summaries omit

The exact regimes above use information not retained by graph and marginal summaries. The Lovász local lemma summarizes event interactions by a dependency graph, and many algorithmic proofs use that graph to organize repair histories. This section shows that the omitted information can change the exact tail and the geometry of the nonterminating tapes of a fixed computation.

6.1 Why dependency graphs enter

Let $\mathcal{A}$ be a finite family of bad events. In the variable model, the standard dependency graph D joins two events when their variable scopes overlap. Shearer's region is the sharp worst-case criterion, based only on D and the marginal probabilities (p_A), for guaranteeing that all bad events can be avoided (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005). Algorithmic local-lemma analyses use the same graph to organize and weigh repair histories (Moser and Tardos, 2010; Haeupler and Harris, 2017).

This compression of an event system is deliberately coarse. It records which repairs may interact, but not set inclusion, higher-order intersections, or the mass of $\bigcup_A A$. The following calculations compare the exact assignment-state description with graph-only bounds and then isolate the information lost by the graph summary.

6.2 The full state space

Every finite variable-model resampling algorithm is a finite-state randomized computation on assignments. Set $P = P_{\text{tape}}$, the sequential source law of the initial assignment and subsequent resampling draws; finite initial factors do not affect the dimensions below. Delete zero-probability outcomes and assume that every surviving source outcome has probability strictly less than one. Fix a deterministic rule for selecting a currently true bad event. On reachable nonterminal assignments and for $\theta \geq 0$, define

$$(B_\theta)_{xy} = \sum_{z\colon F(x,z)=y} \pi_{v(A(x))}(z)^\theta, \tag{57}$$

where $A(x)$ is the selected event and $F(x,z)$ is the updated assignment. Decompose the nonterminal transition graph into reachable strongly connected components C containing a directed cycle, where a strongly connected component is a maximal set of states that are mutually reachable by directed paths. Let $B_{\theta,C}$ be the restriction of B_θ to C. If no such component exists, then $\mathcal{N} = \varnothing$ and its dimension is zero. Otherwise, Proposition 4.1 gives

$$\dim_H^P \mathcal{N} = \max_C \delta_C, \qquad \rho(B_{\delta_C,C}) = 1. \tag{58}$$

Here $\delta_C \in [0,1]$ is the unique solution of the displayed spectral-radius equation for component C. This formula is exact, but its matrix usually has exponentially many rows and columns.

6.3 The graph bound

Classical entropy-compression and Shearer arguments use the dependency graph and the probabilities p_A to bound the total weight of possible repair lists and traces (Moser and Tardos, 2010; Jerrum, 2024; Shearer, 1985; Scott and Sokal, 2005). Only the complete-graph case is needed here. If $D = K_m$ is complete on $m = |\mathcal{A}|$ event vertices, the graph-only enumeration allows every word $(A_1, \ldots, A_t) \in \mathcal{A}^t$ and assigns it weight $\prod_{i=1}^t p_{A_i}$. Its total weight is

$$\left(\sum_{A \in \mathcal{A}} p_A \right)^t .$$

Put $S = \sum_A p_A$ and choose the history law

$$Q_T(A_1, \ldots, A_T) = \frac{\prod_{t=1}^T p_{A_t}}{S^T}$$

on all event words. Theorem 2.1 then gives $\Delta_T = -T \log_2 S$. If $S < 1$, it follows that

$$\mathbb{P}(E_T) \leq S^T. \tag{59}$$

The following dimension estimate requires one common fixed block. Let $\Omega = [q]^b$ carry the uniform law, regard every bad event as a subset of this same block, and suppose that every repair redraws all b coordinates. Thus every repair consumes exactly b q-ary symbols and $|A| = q^b p_A$. Let P be the resulting product source on consumed blocks. Summing the compatible overwritten blocks over all event words gives at most

$$\sum_{(A_1,\ldots,A_T)} \prod_{t=1}^T |A_t| = \left(q^b S \right)^T$$

length-T overwritten-block strings. The final block contributes only the fixed factor q^b, which disappears after normalization. Coding the resulting prefixes gives

$$d_{\text{graph}} = \left[1 + \frac{\log_2 S}{b \log_2 q} \right]_{[0,1]}, \qquad \mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq d_{\text{graph}} \quad (\omega \in \mathcal{N}), \quad \dim_H^P \mathcal{N} \leq d_{\text{graph}}, \tag{60}$$

where $[\cdot]_{[0,1]}$ denotes truncation to $[0, 1]$. These are graph-based upper bounds; they need not equal the survival probability or dimension for a fixed selection rule. Equation (60) is not asserted for unequal repair scopes. Padding shorter repairs would be legitimate only if the padding symbols were declared consumed tape coordinates and included in the metric; no such convention is used here. A short graph description also does not make exact computation easy: evaluating the relevant independence polynomial and testing membership in the general Shearer region are #P-hard in general (Harvey et al., 2018).

6.4 Whole-block resampling

Let Ω be a finite block space with source law π satisfying $0 < \pi(x) < 1$ for every x, and let $\mathcal{A}$ be any finite family of bad subsets. Every repair redraws the whole block from π. Fix a deterministic selector ϕ that, for each $x \in U = \bigcup_{A \in \mathcal{A}} A$, chooses one event $\phi(x)$ containing x. The selector partitions the bad union into the disjoint regions

$$C_A = \{x \in U : \phi(x) = A\}. \tag{61}$$

Let $\mathcal{I} = \{A \in \mathcal{A} : C_A \neq \varnothing\}$ index the nonempty regions. For a laminar family with the inclusion-minimal selector, C_A is A minus its maximal proper subevents. Define

$$w_A(\theta) = \sum_{x \in C_A} \pi(x)^\theta, \qquad W(\theta) = \sum_{A \in \mathcal{I}} w_A(\theta) = \sum_{x \in U} \pi(x)^\theta. \tag{62}$$

Write $w(\theta) = (w_A(\theta))_{A \in \mathcal{I}}$ for the corresponding column vector.

Theorem 6.1 (Exact whole-block calculation)**.** *Assume $U \neq \varnothing$, and let $P = \pi^{\mathbb{N}}$. There is a unique $\delta \in [0, 1]$ such that*

$$W(\delta) = \sum_{x \in U} \pi(x)^\delta = 1. \tag{63}$$

(a) If the finite source π and the set U are computable, then δ and the law Q_δ below are computable, and every nonterminating tape satisfies

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) \leq \delta \qquad (\omega \in \mathcal{N} = U^{\mathbb{N}}). \tag{64}$$

Every tape that is Martin–Löf random for $Q_\delta^{\mathbb{N}}$ satisfies $\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta$. Moreover,

$$\dim_H^P \mathcal{N} = \delta, \qquad \mathbb{P}\{\tau \geq t\} = \pi(U)^t. \tag{65}$$

Under the uniform law on $\Omega = [q]^b$, writing $M = |U|$,

$$\dim_H \mathcal{N} = \delta = \frac{\log_2 M}{b \log_2 q}, \qquad \mathbb{P}\{\tau \geq t\} = \left(\frac{M}{q^b}\right)^t. \tag{66}$$

In particular, under a uniform source, a tape of effective strong dimension greater than δ must make the algorithm terminate.

(b) Group rejected blocks according to the region C_A that contains them. For every $\theta \geq 0$, the resulting weighted transition matrix is

$$\widehat{B}_\theta(A, B) = w_B(\theta) \qquad (A, B \in \mathcal{I}). \tag{67}$$

Its spectral radius is $W(\theta)$, the same as that of the full transition matrix, and the all-ones vector $\mathbf{1} \in \mathbb{R}^{|\mathcal{I}|}$ is a right eigenvector.

(c) The dimension and survival bounds are attained by the following reversible record laws. The dimension-attaining law is

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}. \tag{68}$$

For every T-repair execution, there is a probability record law $G_{\delta,T}$ satisfying

$$\log_2 \frac{G_{\delta,T}(S_T, Y_{1:T})}{P_{\mathrm{tape}}(\Xi_T)} = (1 - \delta) \sum_{t=1}^{T} \log_2 \frac{1}{\pi(Y_t)}. \tag{69}$$

The second is the source conditioned on another rejection, $Q_{\mathrm{surv}}(x) = \pi(x)\mathbf{1}\{x \in U\}/\pi(U)$. The corresponding record law $G_{\mathrm{surv},T}$ satisfies

$$\log_2 \frac{G_{\mathrm{surv},T}(S_T, Y_{1:T})}{P_{\mathrm{tape}}(\Xi_T)} = T \log_2 \frac{1}{\pi(U)}. \tag{70}$$

The proof is given in Appendix D.

Remark 6.2 (The one-state reduction). Every repair redraws the whole block. Consequently, the next step is independent of the current rejected block, and the one-state matrix $[W(\theta)]$ already gives both $W(1) = \pi(U)$ and the dimension root. The disjoint regions retain the recorded reason for rejection, but do not change the stopping dynamics. Partially overlapping local repairs, which leave relevant state unchanged, are the harder case.

The survival law is also a direct application of Theorem 2.1. Relabel each repair by the disjoint region C_A containing the overwritten block; this does not change the run or its resampling rule. Put $r_A = \pi(C_A)$ and choose

$$Q_T(A_1, \ldots, A_T) = \prod_{t=1}^{T} \frac{r_{A_t}}{\pi(U)}.$$

For the relabeled event C_{A_t}, the first term in (6) is $\log_2(1/r_{A_t})$, whereas encoding the region label under Q_T costs $\log_2(\pi(U)/r_{A_t})$ bits. Their difference is $\log_2(1/\pi(U))$ at every repair, giving (70). Conditioning only on the original event A_t would count overlapping mass more than once.

Comparison with the dependency graph. Under the standard variable-overlap definition, all common-block events are pairwise adjacent. The graph calculation therefore uses the one-step weight $\sum_{A \in \mathcal{A}} \pi(A)$, while the true rejection probability is $\pi(U)$. Under the uniform source on $[q]^b$, the resulting upper bound and the true dimension are

$$1 + \frac{\log_2 \sum_{A \in \mathcal{A}} \pi(A)}{b \log_2 q} \quad \text{and} \quad 1 + \frac{\log_2 \pi(U)}{b \log_2 q}, \tag{71}$$

again truncated to $[0, 1]$. Before truncation, the two expressions agree when the events are disjoint, whereas overlap produces overcounting. Truncation can conceal this difference when both values are at least one.

In the variable model, a nonedge certifies disjoint scopes and commuting repairs; an edge marks only possible interaction. A complete graph therefore contains no information about how the corresponding bad events intersect.

Example 6.3 (Pairwise data do not determine nontermination)**.** Let the source be uniform on $\Omega = \{0, 1\}^4$. Begin with four subsets of $\{0, 1\}^3$,

$$A_0 = \{000, 001, 010\}, \quad B_0 = \{000, 011, 100\},$$
$$C_{\triangle,0} = \{001, 011, 101\}, \quad C_{\star,0} = \{000, 101, 110\},$$

and append a zero as the fourth coordinate: $A = A_0 \times \{0\}$, $B = B_0 \times \{0\}$, $C_\triangle = C_{\triangle,0} \times \{0\}$, and $C_\star = C_{\star,0} \times \{0\}$. Compare the whole-block systems

$$\mathcal{S}_\triangle = \{A, B, C_\triangle\}, \qquad \mathcal{S}_\star = \{A, B, C_\star\}.$$

Each event depends essentially on all four coordinates: the displayed three-bit set depends on each of its coordinates, and flipping the appended zero to one changes membership. Thus both event–variable incidence graphs are $K_{3,4}$ and both dependency graphs are K_3.

Every event has probability $3/16$ in both systems. Moreover,

$$A \cap B = \{0000\}, \quad A \cap C_\triangle = \{0010\}, \quad B \cap C_\triangle = \{0110\},$$
$$A \cap B = \{0000\}, \quad A \cap C_\star = \{0000\}, \quad B \cap C_\star = \{0000\}.$$

Hence every pairwise intersection has probability 1/16 in both systems, whereas

$$A \cap B \cap C_\triangle = \varnothing, \qquad A \cap B \cap C_\star = \{0000\}.$$

Inclusion–exclusion gives union sizes six and seven. Therefore, for every deterministic selector,

$$\mathbb{P}_\triangle\{\tau \geq t\} = \left(\frac{3}{8}\right)^t, \qquad \mathbb{P}_\star\{\tau \geq t\} = \left(\frac{7}{16}\right)^t,$$

with survival exponents

$$\alpha_\triangle = \log_2 \frac{8}{3}, \qquad \alpha_\star = \log_2 \frac{16}{7}.$$

Writing $\mathcal{N}_\triangle$ and $\mathcal{N}_\star$ for the two nonterminating sets, Theorem 6.1 gives

$$\max_{\omega \in \mathcal{N}_\triangle} \mathrm{Dim}_{\mathrm{eff}}(\omega) = \dim_H \mathcal{N}_\triangle = \frac{\log_2 6}{4}, \qquad \max_{\omega \in \mathcal{N}_\star} \mathrm{Dim}_{\mathrm{eff}}(\omega) = \dim_H \mathcal{N}_\star = \frac{\log_2 7}{4}.$$

Thus the incidence graph, dependency graph, event marginals, and all pairwise intersection probabilities together determine neither the exact survival exponent nor either tape dimension. The two systems first differ at their triple intersection.

Relation to the variable local lemma. The separation does not contradict the tightness of the Shearer region, which is a worst-case statement over event systems with fixed graph and event probabilities. In fact, the common marginal vector in Example 6.3 satisfies

$$3 \cdot \frac{3}{16} = \frac{9}{16} < 1,$$

and therefore lies strictly inside the Shearer region for K_3. The variable local lemma gives avoidability guarantees from event probabilities and event–variable structure (He et al., 2017); algorithmic refinements give convergence bounds, including criteria that use intersection information (Moser and Tardos, 2010; He et al., 2023). Thus the loss of intersection information by dependency graphs is not itself new. The separation above concerns three different outputs of a fixed process: its survival exponent, maximal effective strong dimension, and Hausdorff dimension. The backward identity supplies tail and individual-tape complexity bounds for every finite run in the model, whether or not a local-lemma criterion is satisfied. For the whole-block systems solved here, combining this identity with the product-set calculation determines the exact survival exponent and both dimensions. What differs in Example 6.3 is not local-lemma feasibility but the exact survival law and exceptional set of the specified whole-block process. This is a tape-level analysis of fixed computations, not a stronger universal local-lemma theorem.

7 Conclusion

Nontermination has two complementary descriptions. Reading a finite run backward gives a likelihood ratio. When the resulting information surplus grows, its tail bounds the probability of continued execution, while its coding form bounds the description length of the consumed tape. Summing the source probabilities of all live prefixes to a variable power gives a pressure: its value at $s = 1$ determines survival, and its critical power gives the Hausdorff dimension of the nonterminating set. The first principle is pathwise; the second is setwise.

Together, these principles give exact answers in two local-repair regimes. For finite extremal partial rejection, source-powered trace enumeration places the nontermination dimension at the

Shearer boundary. For nonextremal disagreement repair, Hamming weight determines every source-power operator, and hence the complete survival law and both tape dimensions, independently of the graph topology and every deterministic nonanticipating repair rule. This calculation uses only $N-1$ pressure states, although exact continuation on a path may require 2^{N-1} states.

The resulting tape-level theory connects running time, individual Kolmogorov complexity, and the geometry of an exceptional set. Its main components are the exact likelihood and KL identity, the source-power extension of extremal trace enumeration, and the all-power reduction for a nonextremal repair process. The whole-block separation identifies a limit of graph summaries: even the dependency graph, all event marginals, and all pairwise intersection probabilities need not determine the exact survival law or either tape dimension, because they omit higher-order intersections.

The central open problem is representation. When the selector is determined by the current assignment, the full assignment together with finite controller memory always gives an exact finite-state operator, but this operator generally has exponentially many states. The next goal is either to compile natural classes of overlapping repairs into polynomial- or fixed-parameter-size source-power states, or to prove that exact survival and dimension are computationally hard from a succinct process description.

A Finite-state computations

Let $\mathcal{G}$ be a finite directed multigraph of nonterminal states with fixed initial state s_0. Each outgoing edge $e : s \to t$ represents a distinct next source outcome and has probability $p(e) > 0$. The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path $w = e_1 \cdots e_n$, traversing states $s_0, s_1, \ldots, s_n$, determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^{n} p(e_i).$$

Let $\mathcal{W}_n$ be the set of surviving paths of length n from s_0. Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies $\max_e p(e) < 1$. For $\theta \geq 0$, define

$$(B_\theta)_{st} = \sum_{e:s\to t} p(e)^\theta, \qquad \Lambda(\theta) = \log_2 \rho(B_\theta). \tag{72}$$

For any square matrix C, its spectral radius is $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$. At $\theta = 1$, B_1 is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity $\Lambda(\theta)$ is the base-two pressure, the exponential growth rate of the total θ-weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension; if there is no such component, the nonterminating set is empty.

Use $P[w]$ as the diameter of the cylinder $[w]$. The resulting Hausdorff dimension is the source-relative dimension $\dim_H^P$ defined in the introduction; for a uniform q-ary source it is ordinary Hausdorff dimension in the q-adic metric. For an infinite path $\omega = e_1 e_2 \cdots$, use the corresponding pointwise definition

$$\mathrm{Dim}_{\mathrm{eff},P}(\omega) = \limsup_{n\to\infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define $\dim_{\mathrm{eff},P}$ by replacing the limsup with a liminf.

Proposition A.1 (Detailed finite-state formula)**.** *There is a unique $\delta \in [0, 1]$ satisfying $\rho(B_\delta) = 1$. Moreover,*

$$\lim_{n\to\infty} \frac{1}{n} \log_2 \sum_{w\in\mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \tag{73}$$

$$\lim_{n\to\infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \tag{74}$$

$$\dim_H^P \mathcal{N} = \delta. \tag{75}$$

If $B_\delta h = h$ with $h_s > 0$ for every state s, then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \tag{76}$$

is a probability law on surviving paths. For every $w = e_1 \cdots e_n \in \mathcal{W}_n$, define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^{n} Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1-\delta)\log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \tag{77}$$

For a computable system with computable Q_δ, every nonterminating tape satisfies $\mathrm{Dim}_{\mathrm{eff},P}(\omega) \le \delta$. *A tape that is Martin–Löf random under Q_δ satisfies* $\dim_{\mathrm{eff},P}(\omega) = \mathrm{Dim}_{\mathrm{eff},P}(\omega) = \delta$.

The requirement that Q_δ be computable is the effectivity condition needed for the pointwise conclusion. In the strict-interior finite cases below it follows automatically: if the edge probabilities are computable and

$$\rho(B_0) > 1 > \rho(B_1),$$

then $\theta \mapsto \rho(B_\theta)$ is uniformly computable and strictly decreasing, so bisection computes δ. Irreducibility makes the Perron eigenvalue simple; solving $B_\delta h = h$ with $\sum_s h_s = 1$ gives a computable positive eigenvector and hence a computable Q_δ. Endpoint roots used below are explicit.

Proof of Proposition A.1. Since the graph contains a directed cycle, $\rho(B_0) \ge 1$. Since B_1 is substochastic, $\rho(B_1) \le 1$. If $\theta' > \theta$, then every nonzero entry contributed by an edge is smaller in $B_{\theta'}$ than in B_θ, because $p(e) < 1$. Strict Perron comparison on the common strongly connected support therefore gives $\rho(B_{\theta'}) < \rho(B_\theta)$. Together with continuity, this proves existence and uniqueness of δ. The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1},$$

where e_{s_0} is the coordinate vector of the initial state and $\mathbf{1}$ is the all-ones column vector. Perron–Frobenius theory proves (73); setting $\theta = 1$ gives (74).

If $\theta > \delta$, then $\rho(B_\theta) < 1$, so the total θ-weight of the length-n surviving cylinders tends to zero. These cylinders cover $\mathcal{N}$, giving $\dim_H^P \mathcal{N} \le \delta$. At $\theta = \delta$, (76) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because h is positive on a finite state space, there is a constant $C < \infty$, independent of w, such that $Q_\delta[w] \le CP[w]^\delta$. The mass-distribution principle states that a probability measure satisfying this cylinder bound forces $\dim_H^P \mathcal{N} \ge \delta$. It therefore gives the required lower bound, proving (75) and (77) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system with computable Q_δ, coding under Q_δ gives, for every surviving prefix w of length n,

$$K(w) \le -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a Q_δ-Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015). □

Under a uniform q-ary source, let $M = (M_{st})$, where M_{st} counts the source symbols that move s to t. Then $B_\theta = q^{-\theta} M$. If

$$\alpha = -\lim_{n\to\infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \ge n\}$$

is the survival exponent, then

$$\dim_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing $h_{\text{surv}} = \log_2 \rho(M)$ for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\text{surv}}, \qquad \dim_H \mathcal{N} = \frac{h_{\text{surv}}}{\log_2 q}.$$

Thus α is the entropy deficit imposed by survival, not an entropy function. Equation (77) is the finite-state form of the comparison used in Theorem 2.1.

B Proof for rejection sampling

Let $p = \pi(U)$, let $\nu = \pi(\cdot \mid U)$ be the law of a rejected symbol, and put $\alpha = \log_2(1/p)$.

Proposition B.1 (Entropy bounds for the rejection dimension)**.** *Assume $0 < p < 1$. If $|U| = 1$, then $\delta = 0$. If $|U| \geq 2$, define the Rényi entropy of order $s \in (0,1)$ by*

$$H_s(\nu) = \frac{1}{1-s} \log_2 \sum_{x \in U} \nu(x)^s,$$

and let $H(\nu)$ be the Shannon entropy (Rényi, 1961; Shannon, 1948). Then

$$\delta = \frac{H_\delta(\nu)}{\alpha + H_\delta(\nu)} \tag{78}$$

and hence

$$\frac{H(\nu)}{\alpha + H(\nu)} \leq \delta \leq \frac{\log_2 |U|}{\alpha + \log_2 |U|}. \tag{79}$$

The two bounds coincide when ν is uniform on U.

Proof of Proposition B.1. For $|U| \geq 2$, the root lies in $(0,1)$. Since $\pi(x) = p\nu(x)$ on U, the defining equation for δ becomes

$$1 = p^\delta \sum_{x \in U} \nu(x)^\delta.$$

Taking logarithms gives $\delta\alpha = (1-\delta)H_\delta(\nu)$, which proves (78). The bounds follow from $H(\nu) \leq H_\delta(\nu) \leq \log_2 |U|$. □

C Proofs for the backward likelihood identity

Proof of Theorem 2.1. Take the ratio of (5) and (4). The final-state factor cancels, as do all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (6).

It remains to use this likelihood ratio. Let $\mathcal{W}_T$ be the prefix-free set of minimal tape prefixes that produce at least T repairs, and let $\mathcal{R}_T$ be the set of formal records on which G_T is defined. For $w \in \mathcal{W}_T$, write

$$R_T(w) = \big(S_T(w), L_T(w), Y_{1:T}(w)\big).$$

The reconstruction argument shows that R_T is injective.

First observe that G_T has total mass at most one. Indeed, for a fixed list $\ell = (A_1, \dots, A_T)$, the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \le 1.$$

Define the corresponding mass on live tape prefixes by

$$\mu_T(w) = G_T(R_T(w)), \qquad w \in \mathcal{W}_T.$$

Injectivity of R_T gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \le \sum_{r \in \mathcal{R}_T} G_T(r) \le 1.$$

Thus μ_T is a subprobability mass function. Its total mass can be strictly less than one because Q_T may itself have mass less than one, or because G_T may assign mass to formal records that are not produced by the algorithm.

For every $w \in \mathcal{W}_T$, the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because $\Delta_T = -\infty$ outside E_T and the cylinders determined by $\mathcal{W}_T$ are disjoint, for every $u \ge 0$,

$$\begin{aligned}
\mathbb{P}\{\Delta_T \ge u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \ge u}} P_{\text{tape}}(w) \\
&= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \ge u}} 2^{-\Delta_T(w)} \mu_T(w) \\
&\le 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \ge u}} \mu_T(w) \\
&\le 2^{-u}.
\end{aligned}$$

This proves (7); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Use the canonical effective encoding of the consumed-choice tree from Section 2.1, and extend μ_T by zero off $\mathcal{W}_T$. Membership in $\mathcal{W}_T$ is decidable uniformly from (T, w) by simulating the deterministic computation, and $w \mapsto R_T(w)$ is uniformly computable. The finite source factors and Q_T are uniformly computable by hypothesis. Hence $(\mu_T)_{T \ge 1}$ is a uniformly lower semicomputable family of subprobability masses (indeed, uniformly computable). The Kraft–Chaitin coding theorem for lower semicomputable semimeasures, equivalently Shannon–Fano coding in this discrete setting (Shannon, 1948; Fano, 1949; Li and Vitányi, 2019), gives

$$K(\Xi_T \mid T, \text{instance}) \le -\log_2 \mu_T(\Xi_T) + O(1).$$

The constant is uniform in T and in the realized run. Since

$$-\log_2 \mu_T(\Xi_T) = -\log_2 P_{\text{tape}}(\Xi_T) - \Delta_T,$$

rearranging proves (8). □

Proof of Corollary 2.2. On $\mathcal{W}_T$,

$$\Delta_T(w) = \log_2 \frac{m_T \nu_T(w)}{p_T P_T(w)}.$$

Taking P_T-expectations gives (9). For the second claim, $\Delta_T = J_T + \log_2 Q_T(L_T)$. If r_T is the conditional law of L_T given E_T, then

$$\sum_{\ell} r_T(\ell) \log_2 Q_T(\ell) = -H_2(r_T) - D_2(r_T \| Q_T)$$

when Q_T has total mass one; allocating less mass can only decrease the left side. The maximum is attained at $Q_T = r_T$. □

Proof of Corollary 2.4. The assumptions give $\Delta_T \geq (\beta - \gamma)T - (C_0 + C_1)$. Equation (14) follows from (7). Equation (8), together with the $O(\log_2 T)$ bits needed to specify T, gives $K(\Xi_T) \leq c_0 + cT - (\beta - \gamma)T + o(T)$ along an infinite execution. If $\beta - \gamma > c$, this is impossible for all large T, so no infinite run exists. Otherwise divide by $c_0 + cT$; the gap between consecutive repair prefixes is the fixed number c of bits. □

D Proof for whole-block resampling

Proof of Theorem 6.1. The function W is continuous and strictly decreasing because $0 < \pi(x) < 1$. Moreover, $W(0) = |U| \geq 1$ and $W(1) = \pi(U) \leq 1$, so (63) has a unique solution in $[0, 1]$.

The effectivity statement is direct. If $|U| = 1$, then $\delta = 0$; if $U = \Omega$, then $\delta = 1$. Otherwise $W(0) > 1 > W(1)$ and W is computable and strictly decreasing, so effective bisection computes δ. The finite law $Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}$ is therefore computable.

For full nonterminal states $x, y \in U$, resampling the common scope gives $B_\theta(x, y) = \pi(y)^\theta$. Hence, for every $x \in C_A$,

$$\sum_{y \in C_B} B_\theta(x, y) = \sum_{y \in C_B} \pi(y)^\theta = w_B(\theta),$$

independently of x. Thus grouping states by the sets C_A leaves the weighted transition calculation unchanged and proves (67). The reduced matrix is $\mathbf{1}\, w(\theta)^\intercal$, so its only nonzero eigenvalue is $W(\theta)$ and $\mathbf{1}$ is a right eigenvector. Proposition 4.1 now gives $\dim_H^P \mathcal{N} = \delta$. Each repair draws a fresh block from π and continues precisely on U, proving the survival formula and its uniform specialization.

For the dimension bound, observe that Q_δ is computable and supported on U. Every $\omega \in U^{\mathbb{N}}$ therefore satisfies

$$K(\omega_{1:n}) \leq -\log_2 Q_\delta^{\mathbb{N}}[\omega_{1:n}] + O(\log_2 n) = \delta\big(-\log_2 P[\omega_{1:n}]\big) + O(\log_2 n).$$

Division by the source information proves (64). If ω is Martin–Löf random for $Q_\delta^{\mathbb{N}}$, Levin–Schnorr gives the reverse inequality up to $O(1)$, and hence equality. This also proves (18).

It remains to construct the record laws. The repair list is determined by the overwritten states because $A_t = \phi(Y_t)$. Reversal recovers the used tape from $(S_T, Y_{1:T})$. The source factorization is

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^{T} \pi(Y_t).$$

Since (63) holds, the record law

$$G_{\delta,T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^{T} \Big[\pi(y_t)^\delta \mathbf{1}\{y_t \in U\}\Big]$$

has total mass one. Dividing by the source factorization proves (69). Similarly, define

$$G_{\mathrm{surv},T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^{T} \frac{\pi(y_t)\mathbf{1}\{y_t \in U\}}{\pi(U)}.$$

This is a probability law, and division by the source factorization proves (70). The region containing y_t determines the recorded rejection label $A_t = \phi(y_t)$. □

E Proof for source-power reduction

Proof of Proposition 4.2. Let C lift a function on $\mathcal{C}$ to a function on $\mathcal{X}$ by $(Cf)(x) = f(\phi(x))$. Condition (20) is exactly $B_s C = C R_s$. Since $C\mathbf{1} = \mathbf{1}$, iteration gives (21). For a nonnegative matrix M, $\|M\|_\infty = \max_x (M\mathbf{1})(x)$. Surjectivity of ϕ and (21) therefore give

$$\|B_s^T\|_\infty = \|R_s^T\|_\infty \quad \text{for every } T.$$

Taking Tth roots proves (22) by Gelfand's formula. The eigenfunction statement follows again from $B_s C = C R_s$. Equation (23) has row sum one, and multiplication along a path telescopes to

$$Q_s[w] = P[w]^s \varrho_s^{-T} \frac{h_s(x_T)}{h_s(x_0)},$$

which is (24). □

F Proofs for extremal local repair

Proof of Theorem 5.1. The classical Cartier–Foata identity gives

$$\sum_{h \in \mathcal{M}(D)} \prod_i z_i^{N_i(h)} = \frac{1}{P_D(\mathbf{z})} \tag{80}$$

as a formal power series (Cartier and Foata, 1969; Jerrum, 2024). For extremal partial rejection sampling, traces are in bijection with resampling transcripts: swapping two consecutive nonadjacent labels swaps repairs on disjoint scopes and changes neither the resampling table nor its frontier. The diamond property also makes the transcript independent of the rule used to choose among currently true events (Guo et al., 2019; Jerrum, 2024).

Fix a trace h. Its resampling blocks and final frontier partition the entries of the consumed resampling table. A halting table with transcript h is obtained by putting a configuration from $\mathcal{B}_i$ in every block labelled i and a satisfying assignment in the final frontier. Conversely, every such choice realizes the transcript. One way to see the converse is to use the Cartier–Foata layers of h and read them backward. Writing a bad configuration into every scope of one layer makes exactly the events of that layer true: their scopes are disjoint; their neighbors are excluded by extremality; and every nonneighbor is unchanged from the next layer. The Cartier–Foata condition says that every event in the next layer is a neighbor of, or equal to, an event in the current layer. Induction from the satisfying frontier proves the claim.

All cells in this partition are independent under the source. After taking the sth power and summing their values, a block labelled i contributes $r_i(s)$ and the final frontier contributes $Z_\Sigma(s)$. Thus a trace h contributes

$$Z_\Sigma(s) \prod_i r_i(s)^{N_i(h)}.$$

Summing first over traces of length T and then over T, and applying (80), proves (29)–(30).

The same partition proves the upper bound in (31), with $C_s = \sum_x \pi(x)^s$: simply allow every full assignment at the current frontier. The lower bound must also respect the fixed sequential selector. Fix a satisfying assignment y and, for each i, a configuration $b_i \in \mathcal{B}_i$. Take any trace of length T and any choice of bad configurations in its blocks, put y in the final frontier, and run the fixed selector. This is a halting tape with that trace. Let i be the event repaired last. Every cell of the final frontier on V_i is drawn by this last repair. Replace just those newly drawn values, from $y|_{V_i}$ to b_i. No earlier state or repair choice changes, while the state after the Tth repair has exactly A_i true: its neighbors are excluded by extremality and its nonneighbors have disjoint scopes and remain false.

This map from the tables with fixed final frontier y to live prefixes is injective: the run identifies its last label i, after which restoring $y|_{V_i}$ recovers the original table. It changes source probability by the factor $\pi_{V_i}(b_i)/\pi_{V_i}(y|_{V_i})$. Consequently the lower bound holds with

$$c_s = \pi(y)^s \min_i \left(\frac{\pi_{V_i}(b_i)}{\pi_{V_i}(y|_{V_i})} \right)^s > 0.$$

This proves (31) without assuming that the selector finishes one Cartier–Foata layer before starting the next.

By Cauchy–Hadamard and (31), the source-power growth of live prefixes is $R(s)^{-1}$. At $s = 1$, extremality and inclusion–exclusion give $Z_\Sigma(1) = \mathbb{P}(\Sigma) = P_D(\mathbf{p}) > 0$. More is true. For any $\mathbf{y}$ with $0 \le y_i \le p_i$, independently retain event A_i with probability y_i/p_i. The retained events are still extremal, their marginals are $\mathbf{y}$, and their common avoidance event contains Σ. Inclusion–exclusion therefore gives

$$P_D(\mathbf{y}) \ge \mathbb{P}(\Sigma) > 0 \qquad (0 \le \mathbf{y} \le \mathbf{p}).$$

Compactness and continuity then give $P_D(u\mathbf{p}) > 0$ for all $u \le 1 + \varepsilon$, for some $\varepsilon > 0$. Thus $R(1) > 1$. Summing the exact halting probabilities at $s = 1$ gives (33). The first assertion of (32) follows; the existence of the exponential-rate limit follows from the standard finite Cartier–Foata automaton.

Every entry of every bad local configuration has source probability strictly between zero and one. Put $c = \max_{i, b \in \mathcal{B}_i} \pi_{V_i}(b) < 1$. If $t > s$, then $r_i(t) \le c^{t-s} r_i(s)$ for every i, and hence $a_T(t) \le c^{(t-s)T} a_T(s)$. The finite Cartier–Foata automaton also shows that the growth $R(s)^{-1}$ is continuous. It is therefore strictly decreasing, is less than one at $s = 1$, and is at least one at $s = 0$ because the trace i^T exists for every T. This proves the existence and uniqueness of δ.

For completeness, the same conclusion can be read directly from the full assignment chain for the fixed selection rule. Its total s-weight after T live repairs is $L_T(s)$; the initial assignment contributes only a fixed finite starting vector. Decompose its finite nonterminal graph into reachable cyclic strongly connected components and apply Proposition 4.1 to each component. Equation (31) says that the largest component pressure is $\log R(s)^{-1}$, so the largest component root is δ. This proves $\dim_H^P \mathcal{N} = \delta$. Every infinite path eventually remains in one of these finitely many components. Under computability, coding in that component gives the upper effective-strong-dimension bound, and a Martin–Löf-random path for a critical component attains δ.

For the pointwise statement, the finite component matrices have computable entries. The strict monotonicity just proved makes the critical root computable by bisection, and the normalized positive Perron vector of a critical irreducible component is computable. Hence its Doob law satisfies the effectivity hypothesis of Proposition 4.1.

Finally, the trace series has nonnegative coefficients. Pringsheim's theorem therefore implies that its first singularity on every positive ray occurs at a positive zero of P_D. At $s = \delta$ that singularity is $u = 1$, which proves (34). Equivalently, the open domain of convergence of (80) consists of the vectors $\mathbf{z}$ for which $P_D(\mathbf{y}) > 0$ for every $0 \le \mathbf{y} \le \mathbf{z}$; this is the standard independence-polynomial description of Shearer's region (Shearer, 1985; Scott and Sokal, 2005). $\square$

Proof of Corollary 5.2. For $0 \le s \le 1$, concavity gives, event by event,

$$p_i^s = \left(\sum_{b \in \mathcal{B}_i} \pi_{V_i}(b) \right)^s \le r_i(s) \le k_i^{1-s} p_i^s.$$

The trace coefficients in (28) are coordinatewise increasing in these weights. Their exponential growth rates are strictly decreasing in s, so comparison at every s brackets their critical roots and proves (36). When the weights are constant across vertices, the trace growth is κ_D times that common weight. Solving the two scalar equations gives (37). $\square$

Proof of Corollary 5.3. Atomicity gives $r_i(s) = p_i^s$. In the equal-probability case, $a_T(s) = p^{sT} a_T(0)$, so its exponential growth is $p^s \kappa_D$. Setting $s = 1$ gives the survival rate and setting $s = \delta$ makes the growth equal to one, proving (39). In general, write $\alpha = -\log_2 \gamma(1)$, where $\gamma(s)$ is the trace growth at weights p_i^s. For $0 \le s \le 1$, each length-T trace satisfies

$$2^{(1-s)\ell_{\min} T} \prod_i p_i^{N_i(h)} \le \prod_i p_i^{s N_i(h)} \le 2^{(1-s)\ell_{\max} T} \prod_i p_i^{N_i(h)}.$$

Taking exponential growth rates and using $\gamma(\delta) = 1$ proves (40). $\square$

G Proofs for disagreement repair

Proof of Theorem 5.4. Let $M(x) = \sum_{v \in V} x_v$. Every selected edge contains one zero and one one. If its redraw contains J ones, then

$$M_{t+1} = M_t - 1 + J, \qquad J \sim \mathrm{Binomial}(2, p). \tag{81}$$

Because G is connected, an assignment is live exactly when $1 \le M \le N - 1$. Conditional on any past with $M_t = k$, the four possible redraws have s-weights $q^{2s}, (pq)^s, (pq)^s, p^{2s}$ and move the count by $-1, 0, 0, +1$, respectively. Induction on T therefore shows directly that the source-power sums grouped by M evolve by (42), even when the selector uses the past. The initial value is $\mathrm{Binomial}(N, p)$, proving (43) and (44). This also proves that the full absorption-time law is independent of G and of the selection rule.

For $1 \le k < N$, the positive function h_s in (51) satisfies

$$\begin{aligned}(R_s h_s)(k) &= q^{2s} h_s(k-1) + 2(pq)^s h_s(k) + p^{2s} h_s(k+1) \\ &= 4(pq)^s c_N^2 h_s(k).\end{aligned}$$

Proposition 4.2 therefore gives (45). Equivalently, R_s is diagonally similar to $(pq)^s$ times the tridiagonal matrix with diagonal 2 and off-diagonal 1; its eigenvalues are

$$4(pq)^s \cos^2\left(\frac{r\pi}{2N}\right), \qquad r = 1, \ldots, N-1. \tag{82}$$

Equations (46) and (47) follow by setting $s = 1$ and by solving $\varrho_s = 1$, respectively.

Applying (23) with h_s gives

$$\frac{P_p(z)^s h_s(k + d(z))}{\varrho_s h_s(k)} = \frac{\sin((k + d(z))\vartheta)}{4 c_N^2 \sin(k\vartheta)}.$$

The right-hand side is independent of s and p, and its sum over the four redraws is one because

$$\sin((k-1)\vartheta) + 2\sin(k\vartheta) + \sin((k+1)\vartheta) = 4c_N^2 \sin(k\vartheta).$$

Moreover, $\nu^*(k)Q^*(11 \mid k) = \nu^*(k+1)Q^*(00 \mid k+1)$; the self-loops need no check. This proves reversibility and (49). (50) is (24). At the root $s = \delta_{N,p}$, endpoint ratios are bounded above and below because $1 \le k < N$. Equation (44) gives the Hausdorff upper bound directly. The global positive law Q^* and the bounded likelihood ratio give the matching mass-distribution lower bound and, under computability, the effective-dimension upper bound and attainment on a Q^*-Martin–Löf-random tape. This argument does not require the full assignment chain for a particular selector to be irreducible.

When p and the selection rule are computable, the explicit formula makes $\delta_{N,p}$ computable, and the entries of Q^* are computable algebraic numbers. Finally, suppose $p = 1/2$ and let $u_k = \mathbb{E}[\tau \mid M_0 = k]$. First-step analysis gives

$$u_0 = u_N = 0, \qquad u_k = 1 + \frac{1}{4}u_{k-1} + \frac{1}{2}u_k + \frac{1}{4}u_{k+1}, \quad 1 \le k < N.$$

The unique solution is $u_k = 2k(N-k)$. Since $M_0 \sim \text{Binomial}(N, 1/2)$,

$$\mathbb{E}[M_0(N - M_0)] = \frac{N(N-1)}{4},$$

and averaging u_{M_0} proves (56). □

Proof of Proposition 5.5. For each $S \subseteq \{1, \dots, N-1\}$, let $x_0^S = 0$ and let $x_i^S = 1$ exactly when $i \in S$. Give every repair the common future redraw 00. The successive repair labels are the elements of S in increasing order, after which the computation halts. Distinct sets therefore have distinct output sequences under the same continuation. If two of the x^S had the same summary, deterministic prediction and updating would give the same output sequence, a contradiction. □

H Adjacent rises: an extremal benchmark

Let $n \ge 1$ and let $X_0, \dots, X_n$ be independent bits with $\mathbb{P}\{X_i = 1\} = p_i \in (0,1)$, and set

$$A_i = \{X_{i-1} = 0,\ X_i = 1\}, \qquad 1 \le i \le n.$$

At each step, any deterministic current-assignment rule may select a true A_i; the repair redraws (X_{i-1}, X_i) from its product marginal. For $\theta \ge 0$, let P be the sequential source law and let B_θ be the full live-state source-power operator. For a pair outcome $z = (z_{i-1}, z_i)$, write

$$P_i(z) = (1-p_{i-1})^{1-z_{i-1}} p_{i-1}^{z_{i-1}} (1-p_i)^{1-z_i} p_i^{z_i}.$$

Define the symmetric tridiagonal matrix K_θ, indexed by $0, \dots, n$, by

$$(K_\theta)_{i-1,i} = (K_\theta)_{i,i-1} = \big[(1-p_{i-1})p_i\big]^{\theta/2}, \qquad (K_\theta)_{ii} = 0. \tag{83}$$

For the state lower bound below, an *outcome-driven state representation* is a map σ from assignments to a finite set together with a status map and an update map U such that the state alone reports live versus halted and

$$\sigma(F(x,z)) = U(\sigma(x), z)$$

whenever the deterministic rule repairs the live assignment x and observes the pair outcome $z \in \{00, 01, 10, 11\}$; here $F(x,z)$ is the resulting assignment. Thus no part of the current assignment is available outside the state.

Theorem H.1 (Adjacent-rise pressure)**.** *For every deterministic current-assignment selection rule and every $\theta \geq 0$,*

$$\rho(B_\theta) = \rho(K_\theta)^2. \tag{84}$$

Consequently,

$$\lim_{T\to\infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \rho(K_1)^2, \qquad \dim_H^P \mathcal{N} = \delta, \qquad \rho(K_\delta) = 1. \tag{85}$$

The root $\delta \in [0,1]$ is unique. If the biases and selection rule are computable, it is also the largest effective strong dimension of a nonterminating tape and is attained by an explicit edgewise product-form law.

For the least-index rule, however, every outcome-driven state representation has at least 2^n states, including the halting state.

The proof is given in Appendix I.

Corollary H.2 (A linear eigenproblem versus exponential continuation state)**.** *For fixed θ, the matrix K_θ has $2n$ nonzero entries and is constructed in linear time. Its largest eigenvalue, and hence $\rho(B_\theta)$, can be approximated to additive error ε using $O(n\log(1/\varepsilon))$ arithmetic operations by Sturm bisection for a symmetric tridiagonal matrix. The dimension δ is then obtained by one-dimensional bisection in θ. This is an arithmetic-operation statement; bit complexity also depends on how the biases are represented and on numerical conditioning. In contrast, the least-index process has 2^n outcome-driven continuation states.*

The proof is given in Appendix I.

Adjacent events A_i and A_{i+1} are mutually exclusive, while nonadjacent events are independent. Thus this is an extremal instance in the terminology of partial rejection sampling: dependent bad events cannot occur simultaneously (Guo et al., 2019; Jerrum, 2024). In this regime, known Cartier–Foata enumeration identifies repair transcripts with traces and gives the generating function $P_D(\mathbb{P}(A_1)z_1, \ldots, \mathbb{P}(A_n)z_n)^{-1}$, where P_D is the signed independence polynomial of the dependency graph. The theorem above uses the same structure at every source power and makes its tape-geometric consequence explicit; it does not claim a new conditional-output theorem or a new trace enumeration.

For the dependency path D, put

$$q_\theta = P_D\big(\mathbb{P}(A_1)^\theta, \ldots, \mathbb{P}(A_n)^\theta\big) = \sum_{I \in \mathcal{I}(D)} (-1)^{|I|} \prod_{i\in I} \mathbb{P}(A_i)^\theta.$$

The tridiagonal determinant recurrence gives $q_\theta = \det(I - K_\theta)$. At $\theta = \delta$ the full determinant vanishes; deleting any event strictly lowers the Perron root, so every proper subfamily still has positive Shearer polynomial. Hence the powered marginal vector reaches the Shearer boundary exactly at $\theta = \delta$ (Shearer, 1985; Scott and Sokal, 2005). The determinant and boundary description are classical. The conclusion here is that this boundary value is also the source-relative dimension of the nonterminating tapes of the sequential resampling computation.

I Proofs for adjacent rises

Proof of Theorem H.1. For a live word $x = x_0 \cdots x_n$, let $a(x)$ be its first zero and $b(x)$ its last one. Then $a(x) < b(x)$, and the sets

$$C_{a,b} = \{1^a 0 u 1 0^{n-b} : u \in \{0,1\}^{b-a-1}\}, \qquad 0 \leq a < b \leq n,$$

partition the live assignments. Every repair can only increase a or decrease b. Hence B_θ is block upper triangular over the intervals $[a,b]$.

Fix one diagonal block and write

$$\alpha_j = (1-p_j)^\theta, \qquad \beta_j = p_j^\theta, \qquad c_i = \sqrt{\alpha_{i-1}\beta_i}.$$

Let η and $v=(v_a,\dots,v_b)>0$ be the Perron value and vector of the weighted path $K_{\theta,[a,b]}$. For $a<j<b$, put

$$t_j = \sqrt{\frac{\alpha_{j-1}\alpha_j}{\beta_j\beta_{j+1}}\frac{v_{j-1}}{v_{j+1}}}, \qquad h(x) = \prod_{j=a+1}^{b-1} t_j^{x_j}. \tag{86}$$

Set $t_a=0$ and $t_b=+\infty$ for the boundary formulas. The Perron recurrence gives, for $a\le j<b$,

$$U_j := \alpha_j + \beta_j t_j = \eta\sqrt{\frac{\alpha_j}{\beta_{j+1}}\frac{v_j}{v_{j+1}}}, \qquad t_i = \frac{U_{i-1}U_i}{\eta^2} \qquad (a<i<b). \tag{87}$$

If the selected rise is i, summing $P_i(z)^\theta h(y)/h(x)$ over the outcomes that retain the endpoints of $[a,b]$ gives

$$\frac{(B_{\theta,[a,b]}h)(x)}{h(x)} = \begin{cases} \alpha_a(\alpha_{a+1}/t_{a+1}+\beta_{a+1}), & i=a+1,\\ U_{i-1}(\alpha_i/t_i+\beta_i), & a+1<i<b,\\ \beta_b U_{b-1}, & i=b.\end{cases}$$

Equations (87) and the two endpoint Perron equations make each expression equal to η^2. The same is immediate when $b=a+1$. Thus $B_{\theta,[a,b]}h=\eta^2 h$, independently of which true rise the rule selects. Since $h>0$, Collatz–Wielandt gives

$$\rho(B_{\theta,[a,b]}) = \rho(K_{\theta,[a,b]})^2.$$

Every proper interval matrix is a proper principal submatrix of the irreducible K_θ, so strict Perron comparison makes $[0,n]$ the unique maximal block. This proves (84); the finite-state pressure formula gives (85). For $n\ge 2$, $\rho(K_0)=2\cos(\pi/(n+2))>1$. On the other hand, from every live word a sequence of at most $n+1$ outcomes 00 removes the right endpoint of one selected rise at each step and reaches a terminal nonincreasing word 1^a0^{n+1-a}. The probability of this sequence is bounded below uniformly over the finite state space. Hence $\rho(B_1)<1$ and therefore $\rho(K_1)<1$. Strict monotonicity of $\rho(K_\theta)$ now gives a unique root in $(0,1)$. For $n=1$, $\rho(K_0)=1$ and $\delta=0$.

The law attaining the dimension is also local. At $\theta=\delta$, use the preceding t_j on $C_{0,n}$ and define

$$U_j=\alpha_j+\beta_j t_j, \qquad V_j=\alpha_j/t_j+\beta_j, \qquad U_{i-1}V_i=1.$$

Here $t_0=0$ and $t_n=+\infty$, interpreted by limits. Thus U_j is used for $0\le j<n$ and V_j for $0<j\le n$. If rise i is selected, redraw its two coordinates independently according to

$$\begin{aligned} L_j(0)&=\alpha_j/U_j, &\qquad L_j(1)&=\beta_j t_j/U_j,\\ R_j(0)&=\alpha_j/(t_jV_j), &\qquad R_j(1)&=\beta_j/V_j,\end{aligned}$$

using $L_{i-1}\otimes R_i$. The boundary marginals are degenerate, so the run remains in $C_{0,n}$. Direct substitution shows that this product law is

$$Q_\delta(z\mid x) = P_i(z)^\delta\frac{h(y)}{h(x)}. \tag{88}$$

Thus along every surviving prefix w in this component,

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1-\delta)\log_2 \frac{1}{P[w]} + \log_2 \frac{h(x_T)}{h(x_0)}. \tag{89}$$

The endpoint term is bounded. The mass-distribution argument proves attainment and the Hausdorff lower bound. For the upper and pointwise bounds, note that along any infinite run the pair (a, b) changes at most $2n$ times and is eventually fixed. Applying the same componentwise coding argument on the final $C_{a,b}$, whose pressure root is at most δ, proves the effective and Hausdorff upper bounds for all nonterminating tapes. When the biases are computable, the finite tridiagonal matrix has computable entries, strict monotonicity makes δ computable by bisection, and its normalized Perron vector is computable. Thus the displayed critical law is computable, as required for the pointwise attainment statement.

It remains to prove the state bound. For each $S \subseteq \{1, \dots, n\}$, let $x_0^S = 0$ and $x_i^S = 1$ exactly for $i \in S$. Under the least-index rule, repeated outcome 00 deletes the leftmost one; this distinguishes sets of different sizes. If $S = \{s_1 < \cdots < s_k\}$ and $T = \{t_1 < \cdots < t_k\}$ are distinct, choose $r < k$ with unequal suffix sums. After r outcomes 00, repeated outcome 10 decreases by one the inversion count

$$I(x) = \#\{(i,j) : i < j,\ x_i = 0,\ x_j = 1\}.$$

The two counts are

$$\sum_{j=r+1}^{k} s_j - \binom{k-r}{2} \quad \text{and} \quad \sum_{j=r+1}^{k} t_j - \binom{k-r}{2},$$

so one run halts first under a common continuation. The 2^n assignments x^S are therefore pairwise continuation-distinct. □

Proof of Corollary H.2. The characteristic polynomials of the leading principal submatrices satisfy the three-term recurrence

$$D_{j+1}(\lambda) = \lambda D_j(\lambda) - (K_\theta)_{j-1,j}^2 D_{j-1}(\lambda).$$

It costs $O(n)$ operations to evaluate the Sturm sequence at a trial λ, and $O(\log(1/\varepsilon))$ trials isolate the largest root in a fixed bounded interval. The remaining claims follow from Theorem H.1 and the strict monotonicity of $\rho(K_\theta)$. □

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